



Muhammed Sarfras P C

Machine Learning Engineer

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Professional Summary

Machine Learning Engineer specializing in NLP, RAG architectures, and Time-Series Forecasting. Proven track record in deploying scalable ML solutions with MLOps best practices (Docker, MLflow, CI/CD). Experienced in transforming complex business requirements into production-ready pipelines with robust monitoring and explainability. Strong mathematical foundation with a focus on data-driven decision making.

Technical Skills

Languages: Python, SQL

ML/DL Frameworks: TensorFlow, PyTorch, Scikit-learn, XGBoost, Transformers (Hugging Face), PEFT, LangChain, Sentence-Transformers

Data & Analytics: Pandas, NumPy, SHAP, LIME, Matplotlib, Seaborn, Tableau, Power BI

MLOps & Deploy: Docker, MLflow, FastAPI, Streamlit, GitHub Actions, GCP, AWS, Model Context Protocol (MCP), Vector Databases, PostgreSQL

Specializations: RAG, LLM Fine-Tuning (LoRA/QLoRA), Time-Series Forecasting, NLP, Computer Vision, Clustering, Explainable AI (XAI), Anomaly Detection

Professional Experience

Assistant Administrative Manager, Britco & Bridco Pvt Ltd –
Delhi, India

Oct 2022 – Oct 2024 | Certificate

- Streamlined cross-departmental operations while pursuing Mathematics Honors, demonstrating exceptional time management.
- Enhanced operational efficiency through data-driven decision making and process optimization.
- Developed analytical dashboards for tracking KPIs and generating actionable business insights.

Professional Projects

Crypto AI Analytics Platform - End-to-End ML System

Live Demo | GitHub

AI-powered analytics platform integrating forecasting, sentiment, and on-chain intelligence in a monolithic architecture.

- Architected a monolithic system using the **Model Context Protocol (MCP)** to orchestrate analytical modules, reducing inter-module latency by **50x** compared to microservice baselines.
- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline with **Qdrant** and fine-tuned **DistilRoBERTa**, achieving a **93.5% faithfulness score** by enforcing citation-grounded responses.
- Developed a forecasting engine (**Prophet, CNN-LSTM, Temporal Fusion Transformer**) and integrated **SHAP** for feature interpretability.

Archie - Generative AI RAG Chatbot for Architecture Students

Live Demo | GitHub

LangChain-powered RAG assistant designed to support Architecture students with coursework and design theory.

- Developed an intelligent study assistant capable of processing 1000+ page PDFs, reducing study time by 50%.
- Implemented advanced chunking and hybrid retrieval strategies, achieving 88% answer accuracy with source citations.
- Deployed highly available cloud infrastructure on AWS handling 100+ daily queries.

Additional Projects

YOLOv8 Object Detection MLOps	GitHub
Dockerized YOLOv8-Nano object detection API with FastAPI, deployed on Railway with CI/CD pipelines.	
NASA C-MAPSS Predictive Maintenance	GitHub
Remaining Useful Life (RUL) estimation for turbofan engines using advanced regression and feature engineering.	
Medical Insurance Costs	GitHub
Machine learning regression model to predict medical insurance costs based on individual health factors.	
Credit Card Customer Clustering	GitHub
Customer segmentation using K-means clustering to identify spending patterns and enable targeted marketing.	
Spam Detector	GitHub
NLP-based system classifying emails as spam or legitimate using Naive Bayes and TF-IDF vectorization.	
Delivery Time Predictor	GitHub
Logistics optimization model predicting delivery times using regression analysis on historical shipping data.	
HR Analytics Dashboard	GitHub
Interactive Tableau dashboard visualizing employee performance, retention rates, and workforce demographics.	
Mall Customer Segmentation	GitHub
Unsupervised learning project grouping mall visitors by income and spending score to optimize retail strategy.	
Sentiment Analysis	GitHub
Natural Language Processing project analyzing text data to determine sentiment polarity using ML classifiers.	

Education

Brotopype — Data Science and AI/ML Program	Oct 2024 – Jan 2025 Certificate
• Comprehensive specialization focused on Machine Learning , Deep Learning , MLOps , and Cloud Deployment .	
BSc Mathematics (Honours) , Delhi University	Oct 2021 – Jun 2024
• Specialization: Mathematics , Statistics , Linear Algebra , Calculus .	

Certifications & Achievements

ML/AI: DeepLearning.AI (Supervised ML) | Simplilearn (RAG)
Data Science: IBM (Data Science) | HP LIFE (Analytics)
Technical: HackerRank (SQL Advanced) | Coursera (Excel)